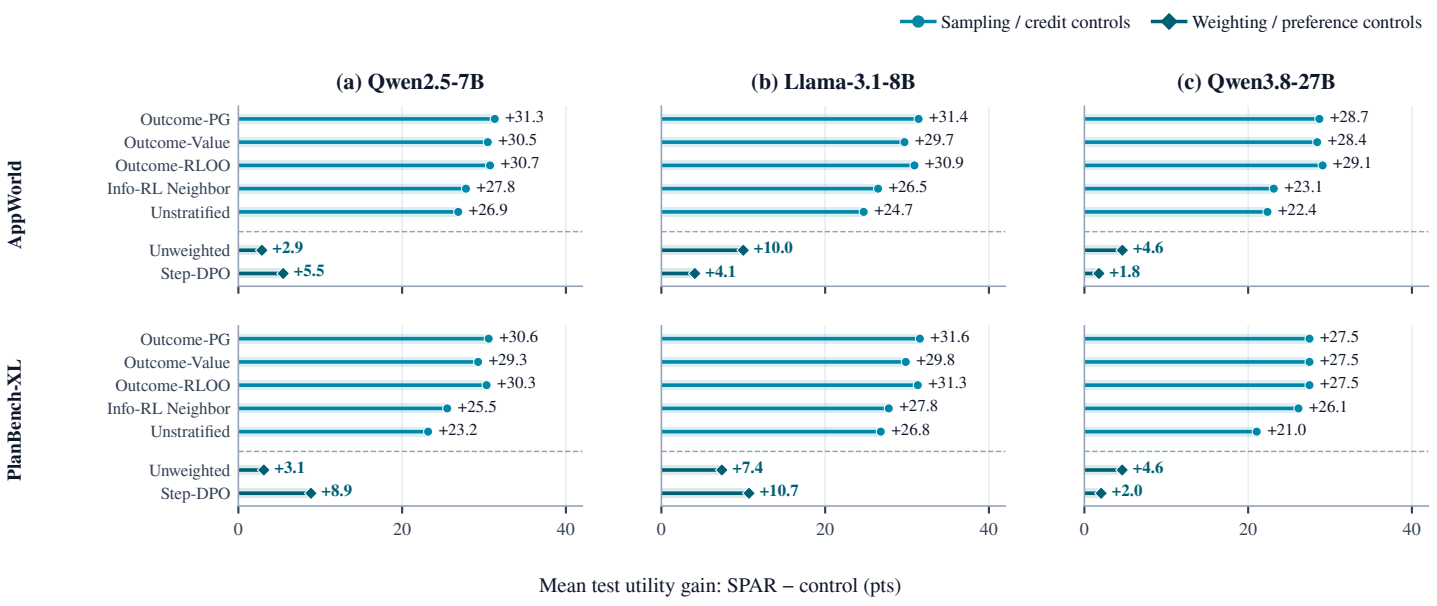
Sampling / credit controls
Weighting / preference controls
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
AppWorld
PlanBench-XL
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
Step-DPO
+31.3
+30.5
+30.7
+27.8
+26.9
+2.9
+5.5
+31.4
+29.7
+30.9
+26.5
+24.7
+10.0
+4.1
+28.7
+28.4
+29.1
+23.1
+22.4
+4.6
+1.8
+30.6
+29.3
+30.3
+25.5
+23.2
+3.1
+8.9
+31.6
+29.8
+31.3
+27.8
+26.8
+7.4
+10.7
+27.5
+27.5
+27.5
+26.1
+21.0
+4.6
+2.0
0
20
40
Mean test utility gain: SPAR − control (pts)